Basic Electric Circuits

Thevenin's and Norton's Theorems

THEVENIN & NORTON

THEVENIN'S THEOREM:

Consider the following:

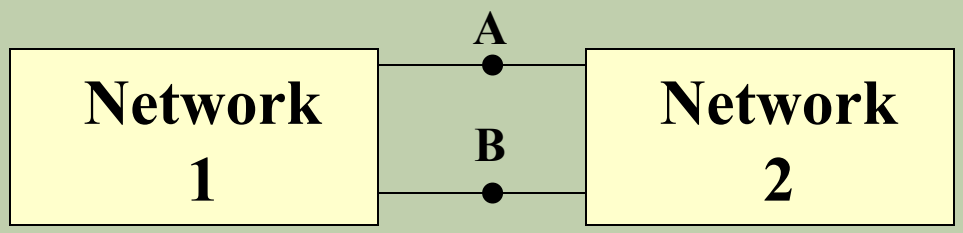


Figure 10.1: Coupled networks.

For purposes of discussion, at this point, we consider that both networks are composed of resistors and independent voltage and current sources

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THEVENIN'S THEOREM:

Suppose Network 2 is detached from Network 1 and we focus temporarily only on Network 1.

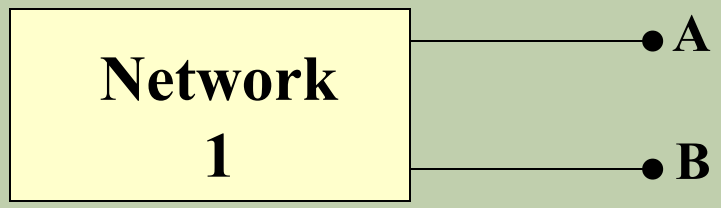
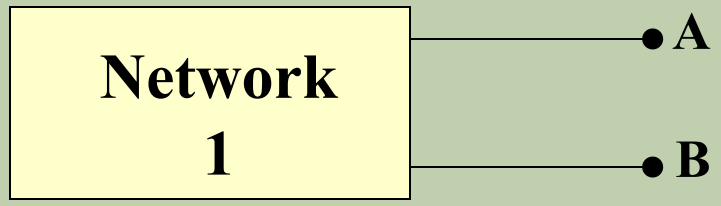


Figure 10.2: Network 1, open-circuited.

Network 1 can be as complicated in structure as one can imagine. Maybe 45 meshes, 387 resistors, 91 voltage sources and 39 current sources.

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THEVENIN'S THEOREM:



Now place a voltmeter across terminals A-B and read the voltage. We call this the open-circuit voltage.

No matter how complicated Network 1 is, we read one voltage. It is either positive at A, (with respect to B) or negative at A.

We call this voltage V_{oc} and we also call it $V_{THEVENIN} = V_{TH}$

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THEVENIN'S THEOREM:

- We now deactivate all sources of Network 1.
- To deactivate a voltage source, we remove the source and replace it with a short circuit.
- To deactivate a current source, we remove the source.

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THEVENIN'S THEOREM:

Consider the following circuit.

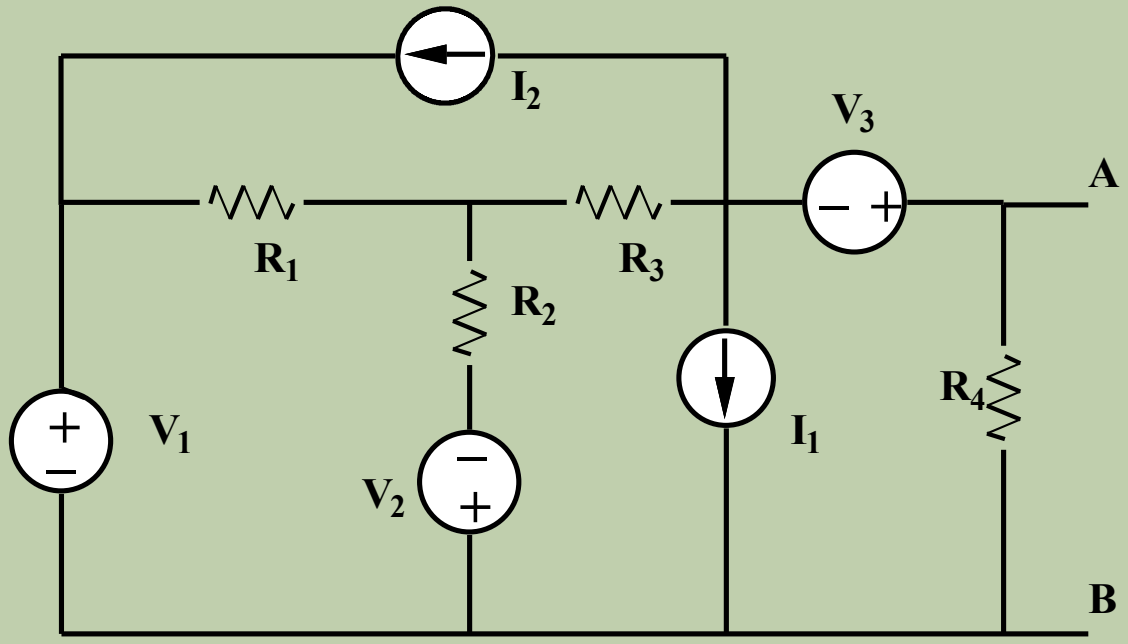


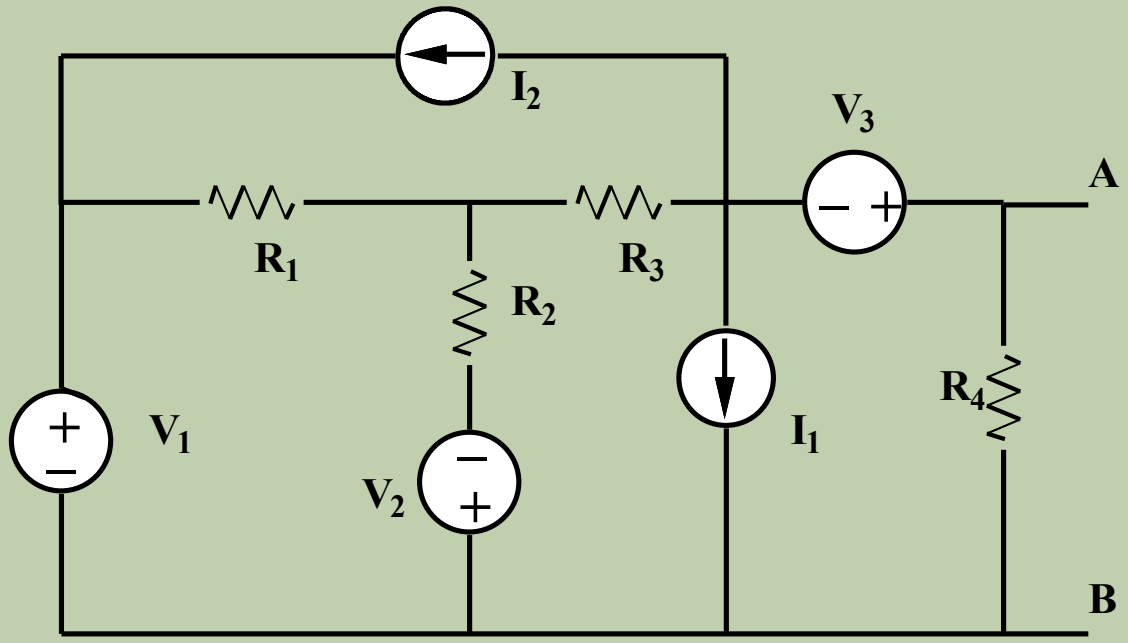
Figure 10.3: A typical circuit with independent sources

How do we deactivate the sources of this circuit?

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THEVENIN'S THEOREM:

Consider the following circuit.



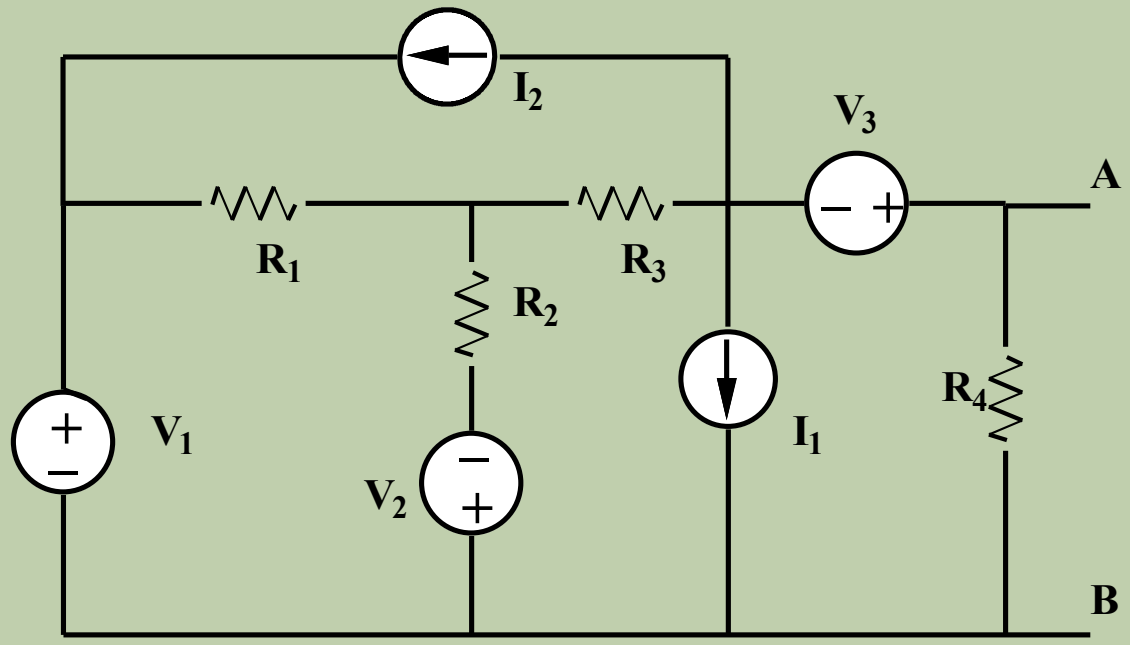
The internal resistance of an ideal voltage source?

The internal resistance of an ideal current source?

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THEVENIN'S THEOREM:

Consider the following circuit.



The internal resistance of an ideal voltage source is zero.

The internal resistance of an ideal current source is infinite.

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THEVENIN'S THEOREM:

When the sources are deactivated the circuit appears as in Figure 10.4.

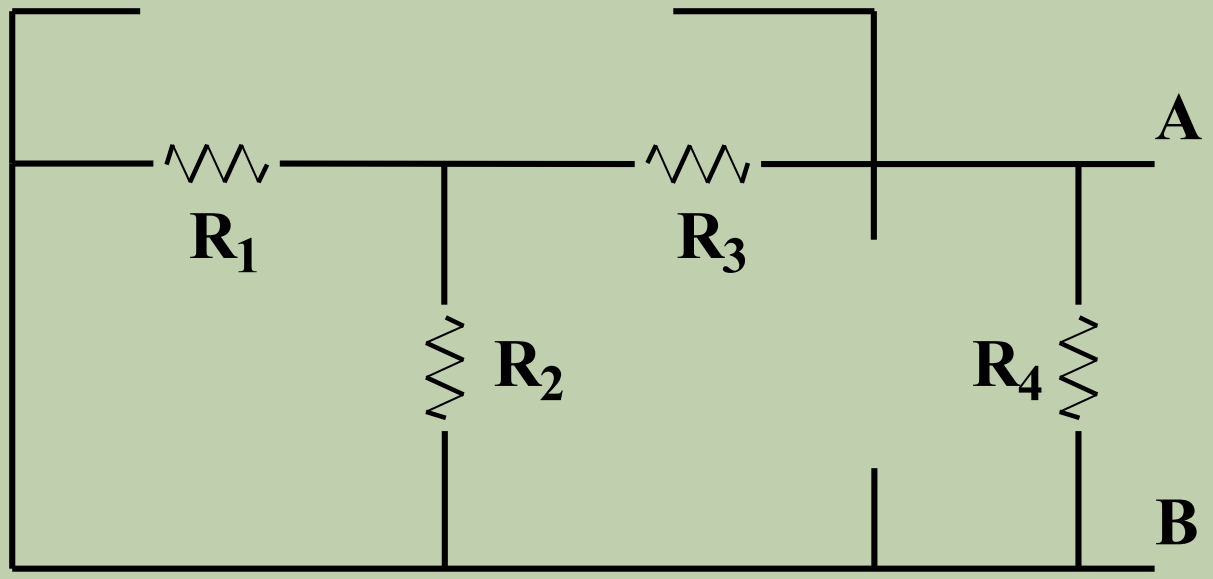


Figure 10.4: Circuit of Figure 10.3 with sources deactivated

Now place an ohmmeter across A-B and read the resistance. If $R_1 = R_2 = R_4 = 20\ \Omega$ and $R_3 = 10\ \Omega$ then the meter reads ?

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THEVENIN'S THEOREM:

We call the ohmmeter reading, under these conditions, $R_{THEVENIN}$ and shorten this to R_{TH}. Therefore, the important results are that we can replace Network 1 with the following network.

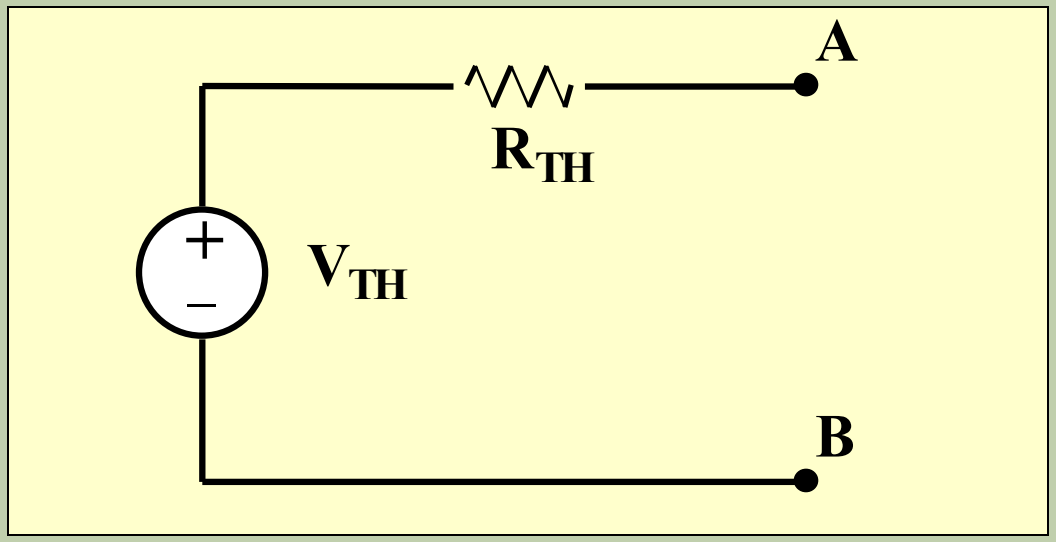


Figure 10.5: The Thevenin equivalent structure.

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THEVENIN'S THEOREM:

We can now tie (reconnect) Network 2 back to terminals A-B.

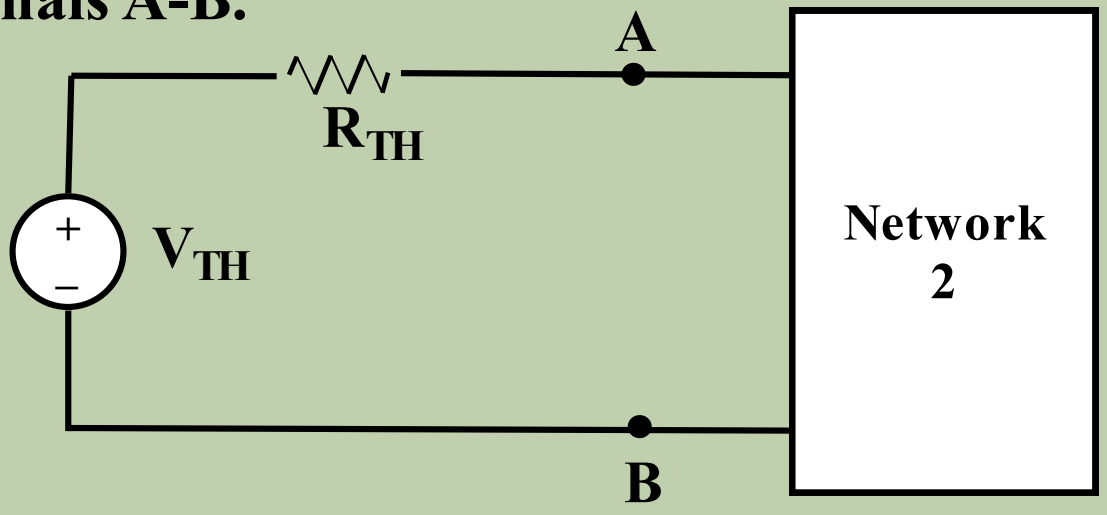


Figure 10.6: System of Figure 10.1 with Network 1 replaced by the Thevenin equivalent circuit.

We can now make any calculations we desire within Network 2 and they will give the same results as if we still had Network 1 connected.

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THEVENIN'S THEOREM:

It follows that we could also replace Network 2 with a Thevenin voltage and Thevenin resistance. The results would be as shown in Figure 10.7.

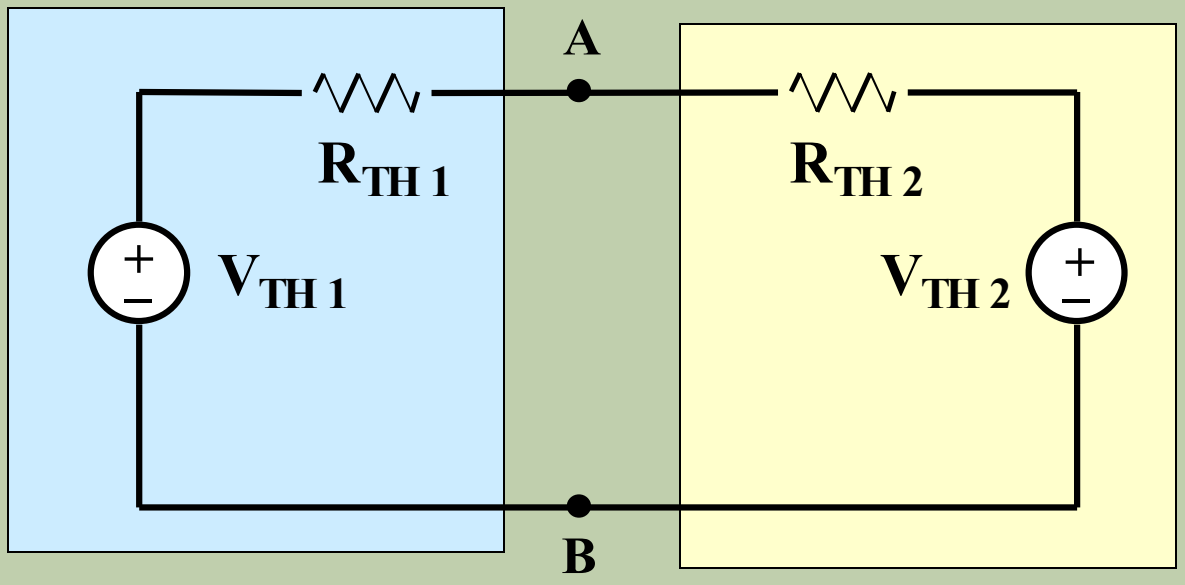


Figure 10.7: The network system of Figure 10.1 replaced by Thevenin voltages and resistances.

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THEVENIN'S THEOREM: Example 10.1.

Find V_X by first finding V_{TH} and R_{TH} to the left of A-B.

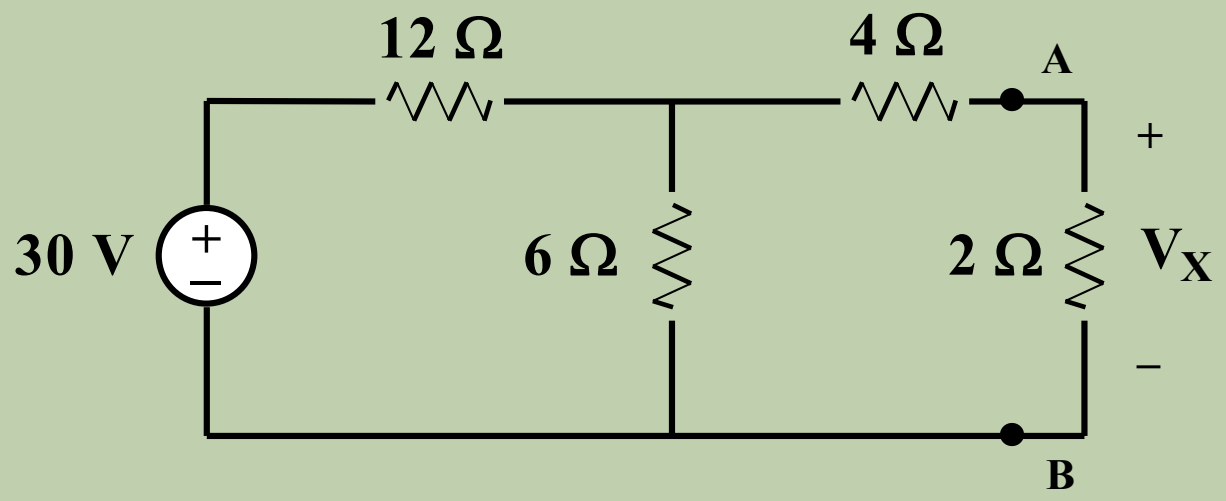


Figure 10.8: Circuit for Example 10.1.

First remove everything to the right of A-B.

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THEVENIN'S THEOREM: Example 10.1. continued

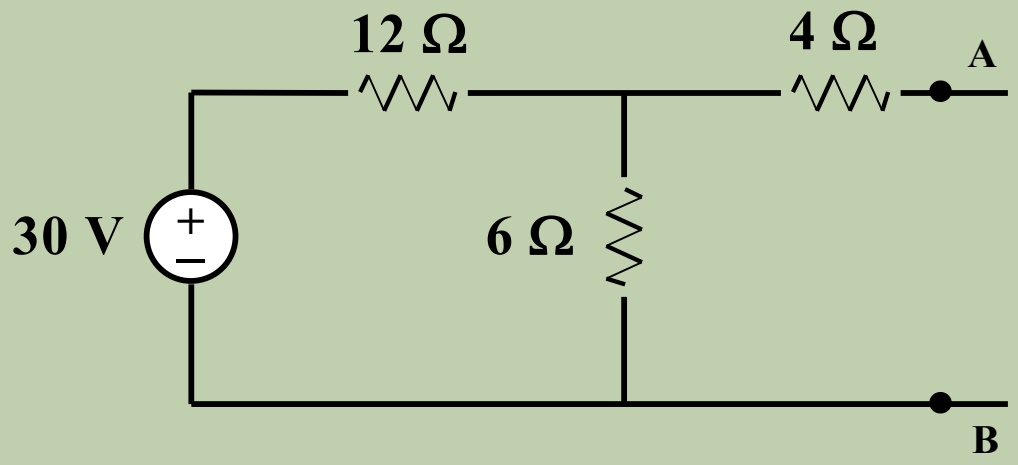


Figure 10.9: Circuit for finding V_{TH} for Example 10.1.

$$V_{AB} = \frac{(30)(6)}{6+12} = 10V$$

Notice that there is no current flowing in the 4 Ω resistor (A-B) is open. Thus, there can be no voltage across the resistor.

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THEVENIN'S THEOREM: Example 10.1. continued

We now deactivate the sources to the left of A-B and find the resistance seen looking in these terminals.

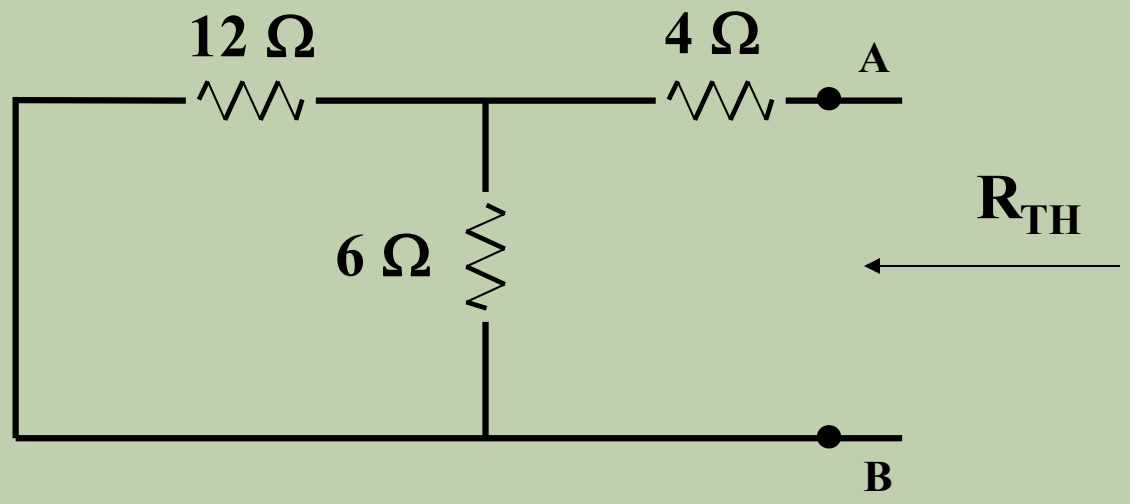


Figure 10.10: Circuit for find R_{TH} for Example 10.10.

We see,

$$R_{TH} = 12||6 + 4 = 8\,\Omega$$

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THEVENIN'S THEOREM: Example 10.1. continued

After having found the Thevenin circuit, we connect this to the load in order to find V_X.

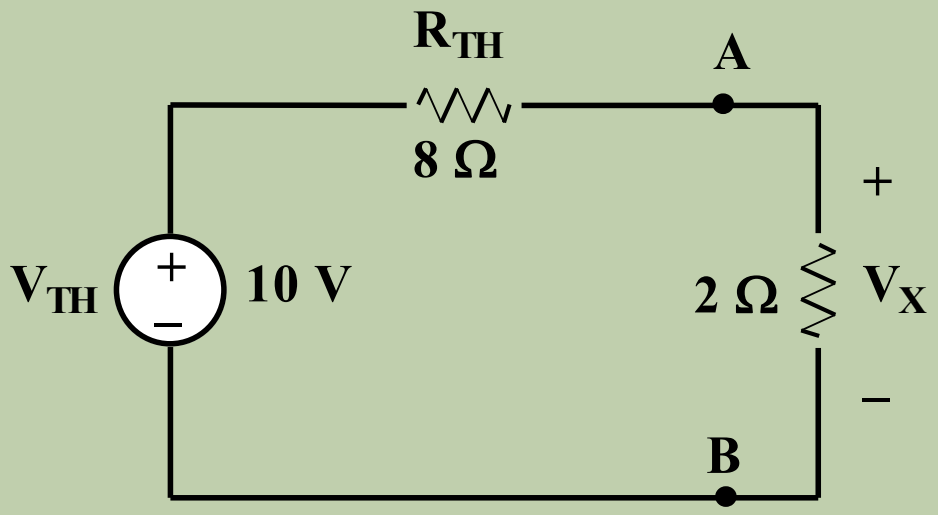


Figure 10.11: Circuit of Ex 10.1 after connecting Thevenin circuit.

$$V_X = \frac{(10)(2)}{2+8} = 2V$$

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THEVENIN'S THEOREM:

In some cases it may become tedious to find R_{TH} by reducing the resistive network with the sources deactivated. Consider the following:

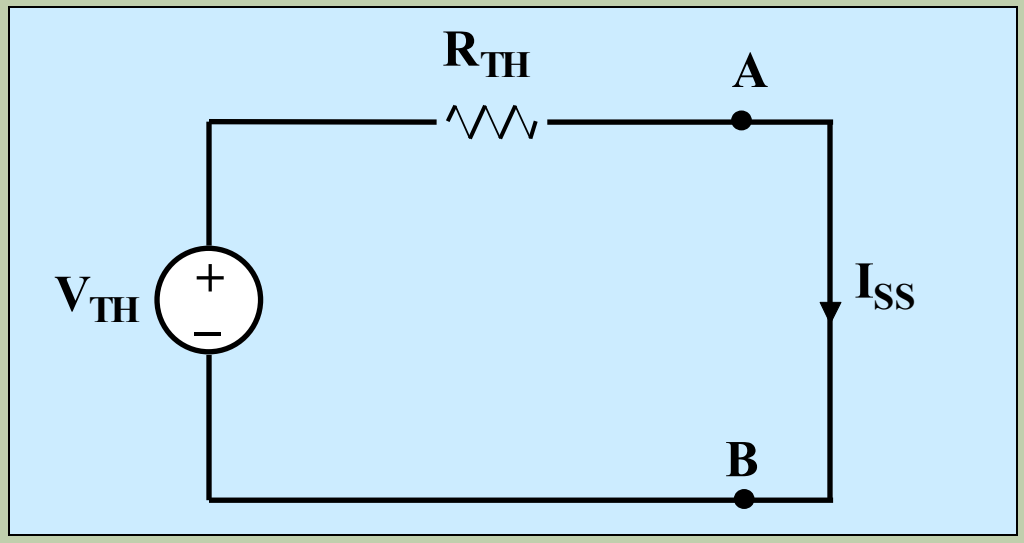


Figure 10.12: A Thevenin circuit with the output shorted.

We see;

$$R_{TH} = \frac{V_{TH}}{I_{SS}}$$

Eq 10.1

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THEVENIN'S THEOREM: Example 10.2.

For the circuit in Figure 10.13, find R_{TH} by using Eq 10.1.

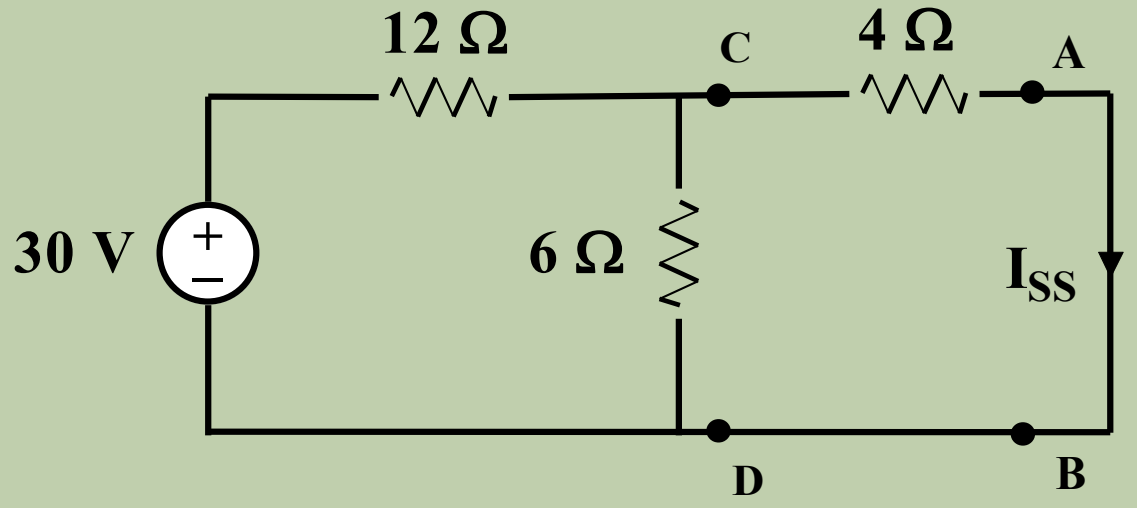


Figure 10.13: Given circuit with load shorted

The task now is to find I_{SS}. One way to do this is to replace the circuit to the left of C-D with a Thevenin voltage and Thevenin resistance.

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THEVENIN'S THEOREM: Example 10.2. continued

Applying Thevenin's theorem to the left of terminals C-D and reconnecting to the load gives,

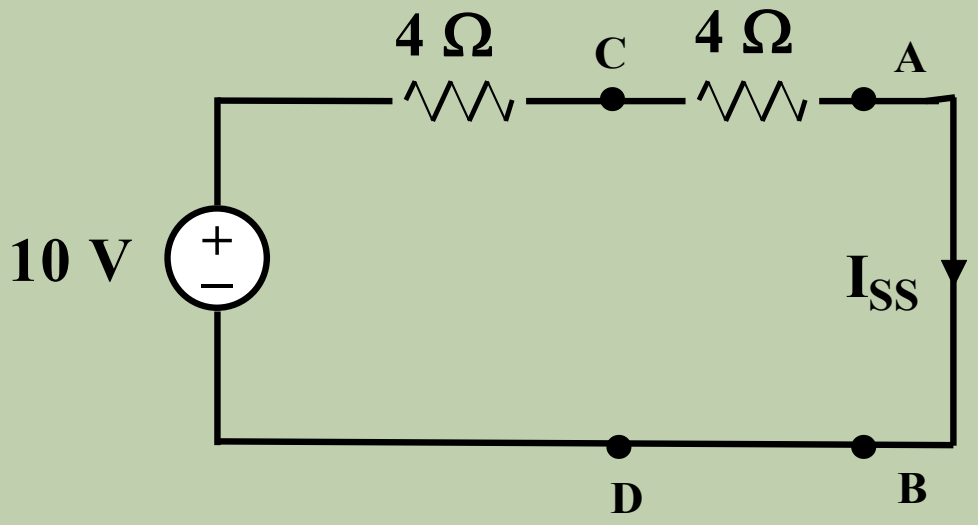


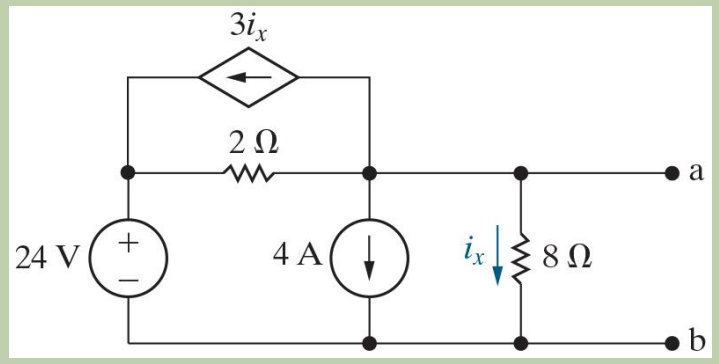
Figure 10.14: Thevenin reduction for Example 10.2.

$$R_{TH} = \frac{V_{TH}}{I_{SS}} = \frac{10}{10/8} = 8\Omega$$

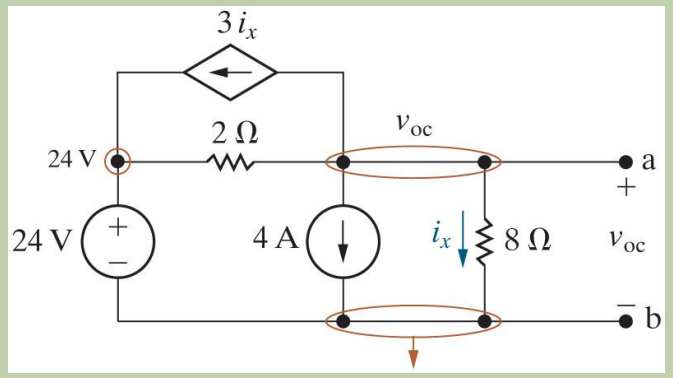
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Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below.



Find V_{oc} using the node voltage method:



At v_{oc}: $\dfrac{v_{oc}-24}{2}+4+\dfrac{v_{oc}}{8}+3i_x=0$

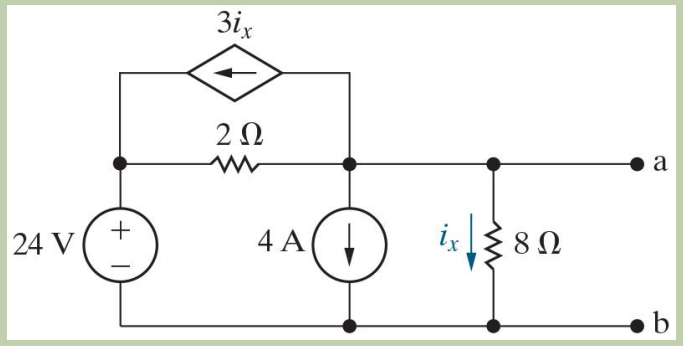
d.s. constraint: $i_x=\dfrac{v_{oc}}{8}$

Solving: $v_{oc}=8\ \text{V}$

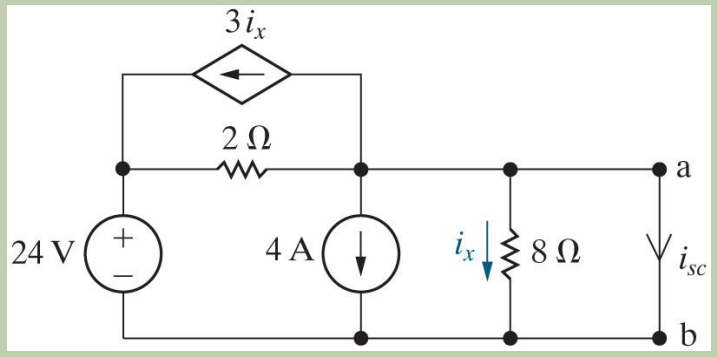
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Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below.



Find i_{sc}:

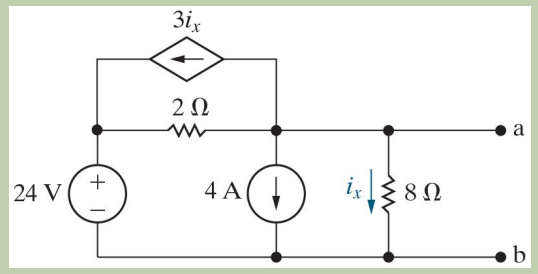


The short circuit eliminates the branch with the 8 Ω resistor from the circuit, which sets $i_x = 0$ and eliminates the dependent source. The simplified circuit is shown and analyzed on the next slide.

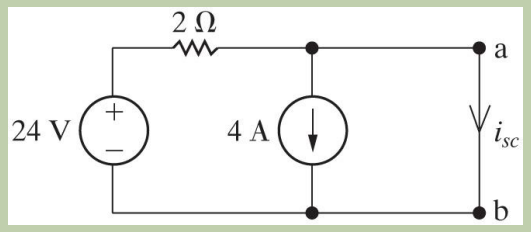
THEVENIN & NORTON

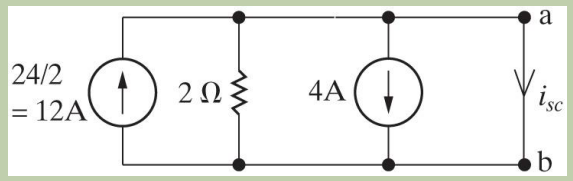
Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below (continued)



Find i_{sc} (continued):





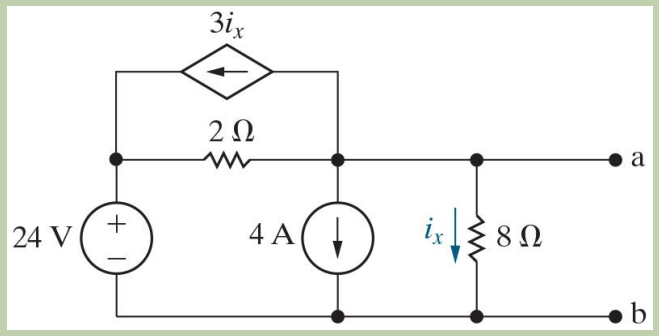
Source transform this circuit to get:

The sum of the currents flows in the short circuit, which bypasses the branch with the 2 Ω resistor. Thus, the short circuit current is 12 – 4 = 8 A.

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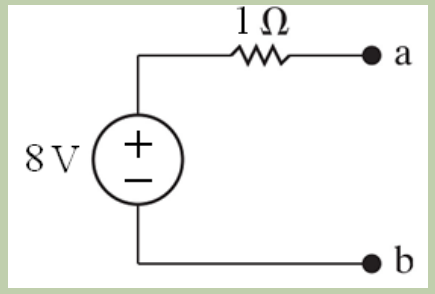
Thevenin and Norton equivalents

- Find the Thevenin equivalent to the left of the terminals a,b in the circuit below (continued)



Find R_{Th} and draw the Thevenin equivalent circuit:

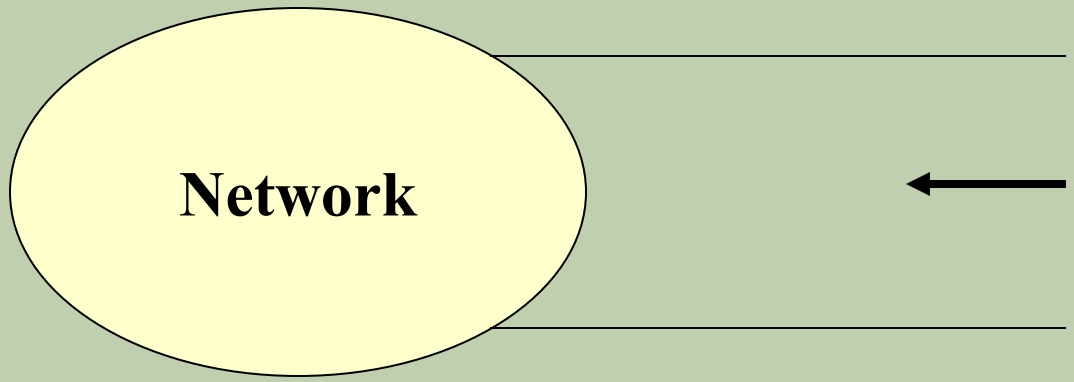
$$R_{Th} = \frac{v_{oc}}{i_{sc}} = \frac{8}{8} = 1\,\Omega$$



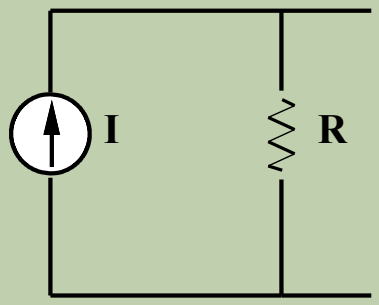
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NORTON'S THEOREM:

Assume that the network enclosed below is composed of independent sources and resistors.



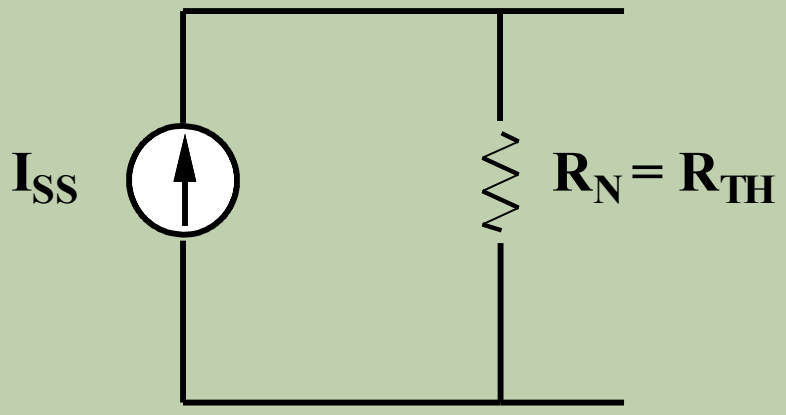
Norton's Theorem states that this network can be replaced by a current source shunted by a resistance R.



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NORTON'S THEOREM:

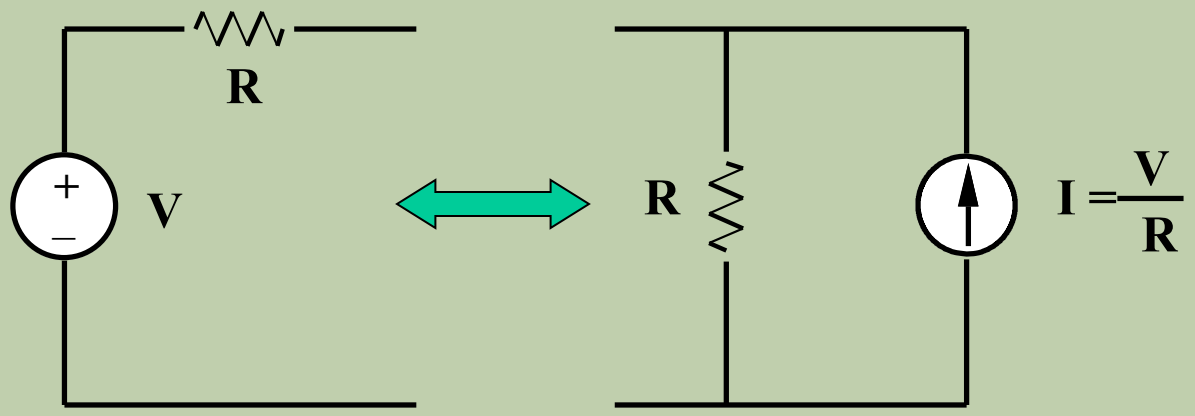
In the Norton circuit, the current source is the short circuit current of the network, that is, the current obtained by shorting the output of the network. The resistance is the resistance seen looking into the network with all sources deactivated. This is the same as R_{TH}.



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NORTON'S THEOREM:

We recall the following from source transformations.



In view of the above, if we have the Thevenin equivalent circuit of a network, we can obtain the Norton equivalent by using source transformation.

However, this is not how we normally go about finding the Norton equivalent circuit.

THEVENIN & NORTON

NORTON'S THEOREM: Example 10.6.

Find the Norton equivalent circuit to the left of terminals A-B for the network shown below. Connect the Norton equivalent circuit to the load and find the current in the 50 Ω resistor.

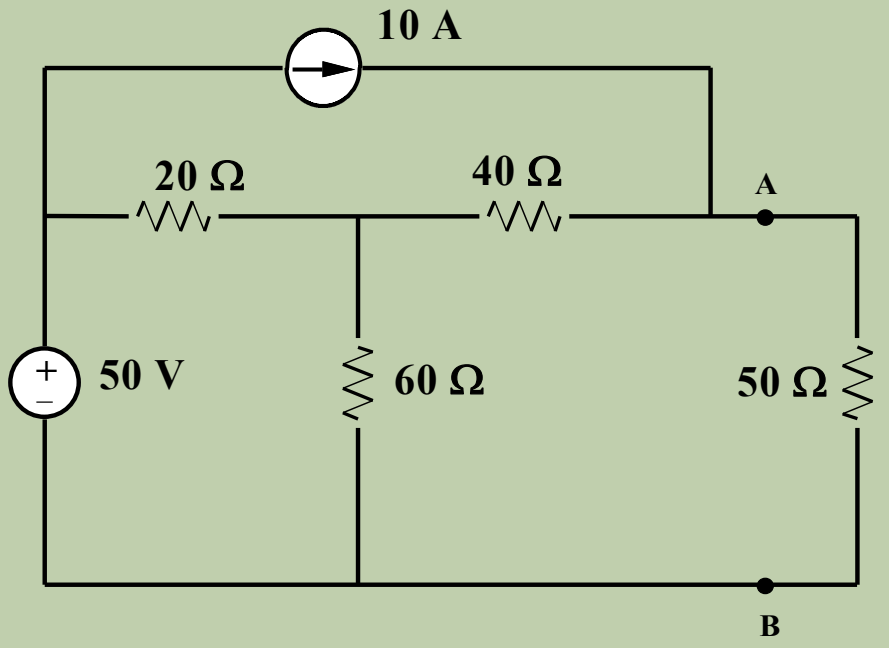


Figure 10.30: Circuit for Example 10.6.

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NORTON'S THEOREM: Example 10.6. continued

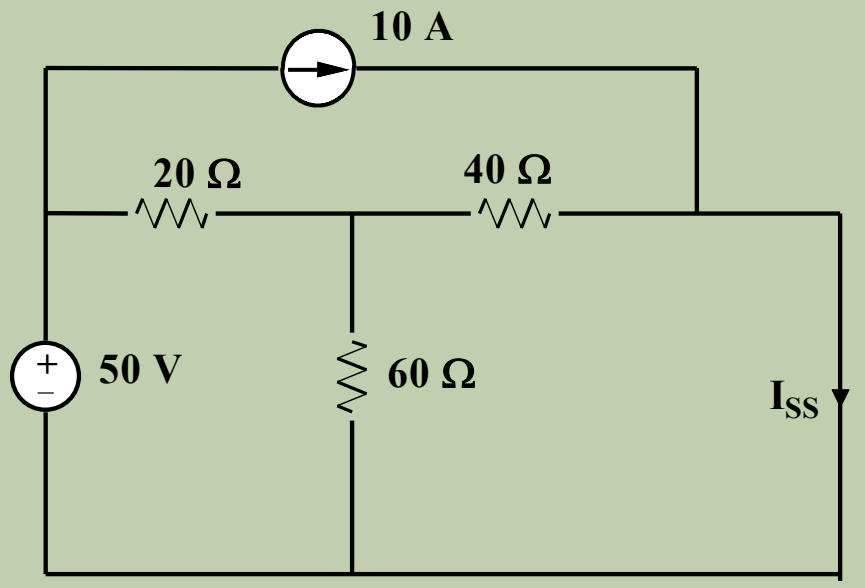


Figure 10.31: Circuit for find I_{NORTON}.

It can be shown by standard circuit analysis that

$$I_{SS}=10.7\ A$$

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NORTON'S THEOREM: Example 10.6. continued

It can also be shown that by deactivating the sources, We find the resistance looking into terminals A-B is

$$R_N = 55\ \Omega$$

R_N and R_{TH} will always be the same value for a given circuit. The Norton equivalent circuit tied to the load is shown below.

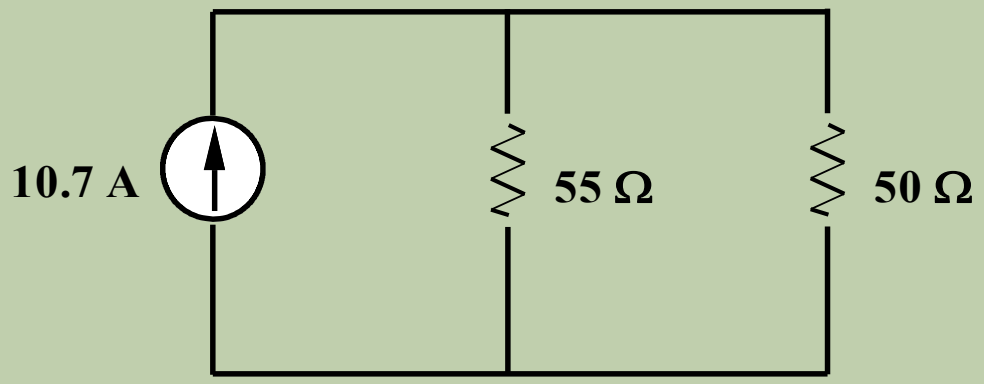


Figure 10.32: Final circuit for Example 10.6.



End of Lecture

Thevenin and Norton